The price of beautiful football

- Does incoming transfer spend accompany better attacking output next season?
- Separate question: does the index track same-season league finish?
- A reproducible data system with measured infrastructure.

Know the source

- 11 pinned CSV tables · 5.74m local rows
- Optional 8,908 completed matches across five leagues
- Headers, hashes and atomic acquisition; raw data excluded from Git
- Boundary: source licence still unconfirmed.

From CSV to the screen

- GET → STORE → PROCESS → SHOW
- PostgreSQL 16 + loader + Python pipeline + FastAPI
- Compose waits for health and successful completion
- Model and measurement artifacts use explicit mounts.

Define the observation before fitting

- Club × season × competition is one observation
- Index = (recorded goals + assists) / source matches
- Target = next consecutive season index − current index
- Exclude inadequate season pairs and competition-unsafe lags.

An honest model comparison

- Earlier seasons train; later seasons test
- Test MAE 0.4856 vs zero-change baseline 0.5114
- About 5% less absolute error; $R^2 = 0.127$
- Separate all-data model powers retrospective scenarios.

See the evidence

- Serie A 24/25 → two clubs → comparison
- Season GF / GA / GD: Inter 79 / 35 / +44; Milan 61 / 43 / +18
- Attack ranks 1 / 3; defense ranks 2 / 7, by per-match goals
- Separate head-to-head: scores 1–2 and 1–1; aggregate goals 2–3
- Open Lab → change one input → reset
- Project page: sample sizes, methods and measurements.
- Shots / shots on target / corners: coverage 8,907 of 8,908 matches
- Poisson model ranks are in-sample; the SOT conversion proxy is not xG

What the data supports

- Transfer spend → next attacking change: r = −0.0036; n = 38,531
- Defender spend → next conceded-goal change: r = +0.0040; p = 0.427
- Same-season index ↔ position: r = −0.793; n = 388
- Different targets; these are associations, not interventions.
- Same temporal split: R^2 0.127 → 0.150 with conceded rate, discipline and net spend
- Red-card difference +0.341 conceded goals; observational, not causal

Measure the whole system

- Three repetitions on Apple M3 Pro / 18 GiB
- Cached-source startup 31.10 s
- 300 lookups: 10.8724 s → 0.0748 s with index
- Database 695.5 MiB; source CSVs 356.6 MiB.

Growth, memory and failure cases

- 25/50/100% of the largest table
- 4× rows: 4.40× load time; 3.35× query time
- Observed per-container memory, not guaranteed instantaneous peak
- 38 regression + 16 live + 15 browser checks.

Boundaries and ownership

- Data quality and causal interpretation stay explicit
- Source licence and independent laptop validation remain open
- Explain the actual work of each real participant
- Take questions on any part of the system.